

# KIRAN TOMLINSON

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## EMPLOYMENT

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|------------------------------------------------------------------------------------------|-------|
| <b>Senior Researcher</b><br><i>Microsoft Research, AI Interaction and Learning Group</i> | 2024– |
| <b>Visiting Instructor</b><br><i>Carleton College, Department of Computer Science</i>    | 2023  |

## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>PhD, Computer Science</b><br>Cornell University<br><i>Advisor: Jon Kleinberg</i>                                                           | 2019–2024 |
| <b>MS, Computer Science</b><br>Cornell University                                                                                             | 2019–2022 |
| <b>BA, Computer Science and Mathematics</b><br>Carleton College<br><i>Summa Cum Laude, Distinction in Computer Science and in Mathematics</i> | 2015–2019 |

## INTERNSHIPS

|                                                                                                           |                     |
|-----------------------------------------------------------------------------------------------------------|---------------------|
| <b>Microsoft Research</b><br><i>Productivity and Intelligence Group, with Jennifer Neville</i>            | Summer 2022         |
| <b>Microsoft</b><br><i>Office of Applied Research, with Longqi Yang and Mengting Wan</i>                  | Summer 2021         |
| <b>NASA Johnson Space Center</b><br><i>Spacecraft Software Engineering Branch, with Nathan Uitenbroek</i> | Summers 2016 & 2017 |

## PUBLICATIONS

\*Equal contribution, ( $\alpha$ – $\beta$ ) alphabetical order

15. Lost in Transmission: When and Why LLMs Fail to Reason Globally  
Tobias Schnabel\*, **Kiran Tomlinson\***, Adith Swaminathan, and Jennifer Neville  
NeurIPS (spotlight, 3%), 2025
14. When the Universe is Too Big: Bounding Consideration Probabilities for Plackett-Luce Rankings  
Ben Aoki-Sherwood, Catherine Bregou, David Liben-Nowell, **Kiran Tomlinson**, and Thomas Zeng ( $\alpha$ – $\beta$ )  
AISTATS, 2025
13. Replicating Electoral Success  
**Kiran Tomlinson**, Tanvi Namjoshi, Johan Ugander, and Jon Kleinberg  
AAAI, 2025
12. The Moderating Effect of Instant Runoff Voting  
**Kiran Tomlinson**, Johan Ugander, and Jon Kleinberg  
AAAI, 2024
11. Graph-based Methods for Discrete Choice  
**Kiran Tomlinson** and Austin R. Benson  
*Network Science* 12(1), 2023
10. Workplace Recommendation with Temporal Network Objectives  
**Kiran Tomlinson**, Jennifer Neville, Longqi Yang, Mengting Wan, and Cao Lu  
KDD, 2023
9. Targeted Training for Multi-organization Recommendation  
**Kiran Tomlinson**, Mengting Wan, Cao Lu, Brent Hecht, Jaime Teevan, and Longqi Yang  
*ACM Transactions on Recommender Systems* 1(3), 2023
8. Ballot Length in Instant Runoff Voting  
**Kiran Tomlinson**, Johan Ugander, and Jon Kleinberg  
AAAI, 2023

7. Choice Set Confounding in Discrete Choice  
**Kiran Tomlinson**, Johan Ugander, and Austin R. Benson  
KDD, 2021
6. Learning Interpretable Feature Context Effects in Discrete Choice  
**Kiran Tomlinson** and Austin R. Benson  
KDD, 2021
5. Choice Set Optimization Under Discrete Choice Models of Group Decisions  
**Kiran Tomlinson** and Austin R. Benson  
ICML, 2020
4. Summarizing Diverging String Sequences, with Applications to Chain-Letter Petitions  
Patty Commins, David Liben-Nowell, Tina Liu, and **Kiran Tomlinson** ( $\alpha$ - $\beta$ )  
31st Annual Symposium on Combinatorial Pattern Matching (CPM), 2020
3. Distance Measures for Tumor Evolutionary Trees  
Zach DiNardo\*, **Kiran Tomlinson**\*, Anna Ritz, and Layla Oesper  
*Bioinformatics* 38(7), 2020
2. Parameter, Noise, and Tree Topology Effects in Tumor Phylogeny Inference  
**Kiran Tomlinson** and Layla Oesper  
*BMC Medical Genomics* 12(10), 2019
1. Examining Tumor Phylogeny Inference in Noisy Sequencing Data  
**Kiran Tomlinson** and Layla Oesper  
IEEE Int. Conf. on Bioinformatics and Biomedicine (BIBM), 2018

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#### PREPRINTS

2. Working with AI: Measuring the Applicability of Generative AI to Occupations  
**Kiran Tomlinson**, Sonia Jaffe, Will Wang, Scott Counts, and Siddharth Suri  
arXiv, 2025
1. Exclusion Zones of Instant Runoff Voting  
**Kiran Tomlinson**, Johan Ugander, and Jon Kleinberg  
arXiv, 2025

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#### PATENTS

2. Intelligent system and method of optimizing cross-team information flow  
Longqi Yang, Mengting Wan, Cao Lu, Jennifer Neville, and **Kiran Tomlinson**  
US20240354703A1 (pending), filed Apr 20, 2023
1. Targeted training of inductive multi-organization recommendation models for enterprise applications  
**Kiran Tomlinson**, Longqi Yang, Mengting Wan, Cao Lu, Brent Hecht, and Jaime Teevan  
US11983649B2, May 14, 2024 (filed Oct 26, 2021)

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#### INVITED TALKS

- “Working with AI: Measuring the Applicability of Generative AI to Occupations.” Louisiana State Board of Elementary and Secondary Education, Sep 2025.
- “Algorithmic Perspectives on Instant Runoff Voting.” Cornell CS Theory Seminar, Aug 2023.
- “Choices in Networks.” NetSci-SINM, Jul 2023.
- “A Computational Look at Ranked-Choice Voting.” Carleton College CS Tea, May 2023.
- “Choice Models for Networks and Networks for Choice Models.” Network Dynamics and Choice Theory Workshop @ UVM, May 2022.
- “Feature Context Effects and Choice Set Confounding in Discrete Choice.” WPI, CS 525, Apr 2021.
- “Feature Context Effects and Choice Set Confounding in Discrete Choice.” Cornell/MIT MURI, Mar 2021.

“Learning Interpretable Feature Context Effects in Discrete Choice.” Stanford Ugander Lab, Dec 2020.

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#### WORKSHOP TALKS

“Working with AI: Measuring the Applicability of Generative AI to Occupations.” Wharton Business & Generative AI Conference, Sep 2025.

“Exclusion Zones in Instant Runoff Elections.” AMS Special Session on Mathematics of Decisions, Elections, and Games @ JMM, Jan 2025.

“Workplace Recommendation with Temporal Network Objectives.” NetSci, Jul 2023.

“Learning Context Effects in Triadic Closure.” NetSci–SINM, Sep 2020.

“Distance Measures for Tumor Evolutionary Trees.” RECOMB–CCB (with Zach DiNardo), May 2019.

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#### AWARDS

|                                                                   |                  |
|-------------------------------------------------------------------|------------------|
| <b>The Web Conference Outstanding Reviewer</b>                    | 2024             |
| <b>Phi Beta Kappa</b>                                             | 2019             |
| <b>Sigma Xi</b>                                                   | 2019             |
| <b>CRA Outstanding Undergraduate Researcher Honorable Mention</b> | 2018             |
| <b>Goldwater Scholarship Honorable Mention</b>                    | 2018             |
| <b>Patricia V. Damon Scholarship</b>                              | 2018             |
| <b>NSF Student Travel Award (to BIBM)</b>                         | 2018             |
| <b>Carleton College Dean’s List</b>                               | 2016, 2017, 2018 |
| <b>National Merit Scholarship</b>                                 | 2015–2018        |

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#### TEACHING

##### **Carleton College (Instructor)**

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| CS 201: Data Structures                 | Winter 2023 ( $\times 2$ ), Spring 2023 |
| CS 202: Mathematics of Computer Science | Spring 2023                             |

##### **Cornell University (Teaching Assistant)**

|                                                          |             |
|----------------------------------------------------------|-------------|
| CS 2110: Object-Oriented Programming and Data Structures | Fall 2019   |
| CS 2800: Discrete Structures                             | Spring 2020 |

##### **Carleton College (Prefect)**

|                                         |                          |
|-----------------------------------------|--------------------------|
| CS 201: Data Structures                 | Winter 2018, Spring 2019 |
| CS 202: Mathematics of Computer Science | Fall 2017, Spring 2018   |
| CS 254: Computability and Complexity    | Fall 2018, Winter 2019   |

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#### SERVICE

##### **Workshop Program Chair**

*What Can(t) Transformers Do? @ NeurIPS ’25*

*The Workshop on Human and Machine Decisions @ NeurIPS ’21*

##### **Program Committee**

*AAAI ’23, ’24, ’25, ’26*

*AISTATS ’25, ’26*

*ICLR ’26*

*ICML ’25*

*ICWSM ’25*

*NeurIPS ’24, ’25*

*WebConf ’24, ’25, ’26*

##### **Journal Reviewer**

*Science Advances*

*Biometrika*

*Data Mining and Knowledge Discovery*

*Annals of Mathematics and Artificial Intelligence*

*Information Processing and Management*

**PhD Admissions Committee**

*Cornell University Computer Science Department*

2021, 2022

**Graduate and Professional Student Assembly Representative (elected)**

*Cornell Computer Science Graduate Organization*

2019–2020